Biochemistry Cume March 31, 2017

Bovine ATP Synthase consists of an F_1 subunit with the composition $\alpha_3\beta_3\gamma\delta\varepsilon$ and an F_o subunit with the composition ab_2c_8. The F_o subunit spans the inner mitochondrial membrane, while the F_1 subunit is located in the matrix of the mitochondria. Vesicles containing bovine ATP synthase (F_oF_1) were prepared by sonicating a mixture of phospholipids, ATP synthase, and gramicidin A in a pH 6.0 buffer containing 3 mM Na^+Cl^-. This procedure results in the incorporation of ATP synthase into the vesicle with the F_1 subunit responsible for the synthesis of ATP oriented toward the outside of the vesicle. The vesicles were then washed with a pH 8.0 buffer containing 300 mM Na^+Cl^-. After this wash, the vesicles thus contain 3 mM Na^+Cl^- and pH 6.0 buffer on the inside of the vesicle, and 300 mM Na^+Cl^- and pH 8.0 buffer on the outside of the vesicle.

A. (30) Discuss the mechanism by which ATP synthase couples the energy of proton transport to ATP synthase. Discuss the roles of each of the components in the F_o and F_1 subunits. How many protons must be transported across the membrane to synthesize each molecule of ATP?

B. (20) Calculate the voltage difference between the outside and inside of the vesicle membrane, assuming that the gramicidin A in the membrane allows efficient and selective transport of Na^+ across the membrane.

C. (20) Calculate the free energy ΔG for transporting 1 mole of protons across the membrane from inside to outside of the vesicle under the above conditions.

D. (15) Calculate the actual free energy ΔG for synthesizing 1 mole of ATP in the solution on the outside of the vesicle, assuming that the outside solution contains the following concentrations: $[ATP] = [P_i] = 20$ mM, $[ADP] = 1.0$ mM.

E. (15) Determine whether the free energy of proton transport by ATP synthase is sufficient to drive the synthesis of ATP by ATP synthase under the conditions given above.

Possibly Useful Equations:

$\Delta G^{o'}$(ATP synthesis) = + 30.5 KJoule/mol.

$\Delta G(X_{out} \rightarrow X_{in}) = RT\ln([X_{in}]/[X_{out}]) + ZF(V_{in} - V_{out})$.

$\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$.

R = 8.3 Joule/degree.mol, F = 96.485 KJoule/volt.mol, T = 298 °K